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README.md	Update README.md	1 minute ago
dipex003.ipynb	Add files via upload	3 days ago

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Histogram Equalization Using OpenCV (Grayscale & Color Images)

Aim

To write a Python program using OpenCV to perform histogram equalization on both grayscale and color images to enhance image contrast and brightness.

The program performs the following operations:

- Read and display a grayscale image
- Plot histogram of the grayscale image
- Apply histogram equalization on grayscale image
- Read and display a color image
- Plot histogram of B, G, R channels
- Convert image to HSV color space
- Apply histogram equalization on the Value (V) channel

- Convert the enhanced image back to BGR format
- Display original and enhanced images with histograms

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (`cv2`)
- NumPy
- Matplotlib

Algorithm

Step 1:

Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2:

Read the image `parrot.jpg` in grayscale format.

Step 3:

Display the grayscale image and plot its histogram.

Step 4:

Apply histogram equalization using `cv2.equalizeHist()` to enhance contrast.

Step 5:

Display original grayscale image, its histogram, enhanced image, and its histogram using a 2 × 2 grid.

Step 6:

Read the same image in color format.

Step 7:

Split the image into B, G, R channels and plot their histograms.

Step 8:

Convert the image from BGR to HSV color space.

Step 9:

Apply histogram equalization on the V (Value) channel.

Step 10:

Merge the channels and convert the image back to BGR format.

Step 11:

Display original color image, histogram, enhanced image, and enhanced histogram using a 2 × 2 grid.

Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image in grayscale
img = cv2.imread('saveetha.jpg', cv2.IMREAD_GRAYSCALE)

# Display original grayscale image
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')
plt.show()

# Display original histogram
plt.hist(img.ravel(), 256, range=[0, 256])
plt.title('Original Image Histogram')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')
plt.show()

# Histogram Equalization for grayscale image
img_eq = cv2.equalizeHist(img)

# Display equalized histogram
plt.hist(img_eq.ravel(), 256, range=[0, 256])
plt.title('Equalized Histogram')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')
plt.show()

# Display equalized grayscale image
plt.imshow(img_eq, cmap='gray')
plt.title('Equalized Image')
plt.axis('off')
plt.show()


# Read image in color
img = cv2.imread('saveetha.jpg', cv2.IMREAD_COLOR)

# Convert BGR image to HSV
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

```
# Equalize the V (Value) channel
img_hsv[:, :, 2] = cv2.equalizeHist(img_hsv[:, :, 2])

# Convert HSV back to BGR
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)

# Display color equalized image
plt.imshow(img_eq[:, :, ::-1])
plt.title('Equalized Color Image')
plt.axis('off')
plt.show()

# Display histogram of equalized color image
plt.hist(img_eq.ravel(), 256, range=[0, 256])
plt.title('Histogram Equalized')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')
plt.show()


# Compare original and equalized color images
plt.figure(figsize=(20, 10))

plt.subplot(2, 2, 1)
plt.imshow(img[:, :, ::-1])
plt.title('Original Color Image')
plt.axis('off')

plt.subplot(2, 2, 2)
plt.imshow(img_eq[:, :, ::-1])
plt.title('Equalized Image')
plt.axis('off')

plt.show()


# Compare original and equalized histograms
plt.figure(figsize=(15, 4))

plt.subplot(1, 2, 1)
plt.hist(img.ravel(), 256, range=[0, 256])
plt.title('Original Image Histogram')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')

plt.subplot(1, 2, 2)
plt.hist(img_eq.ravel(), 256, range=[0, 256])
plt.title('Histogram Equalized')
plt.xlabel('Pixel Intensity')
plt.ylabel('Frequency')

plt.show()
```

Developed By:

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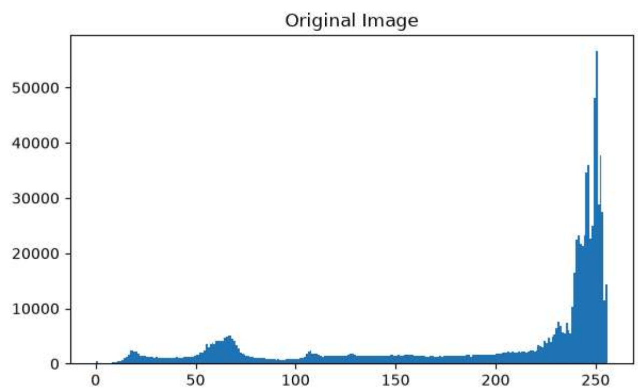
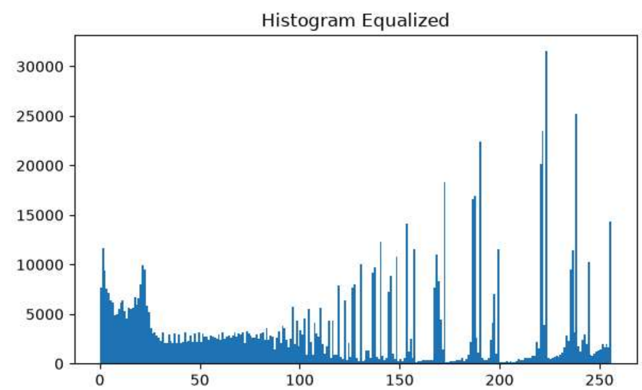
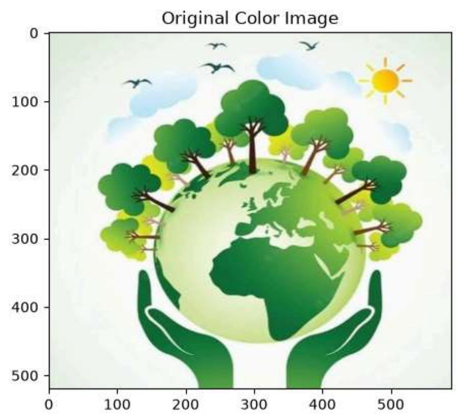
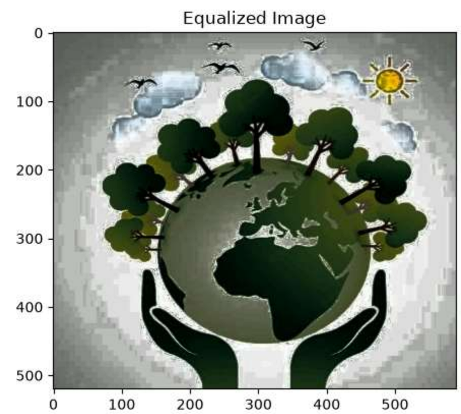
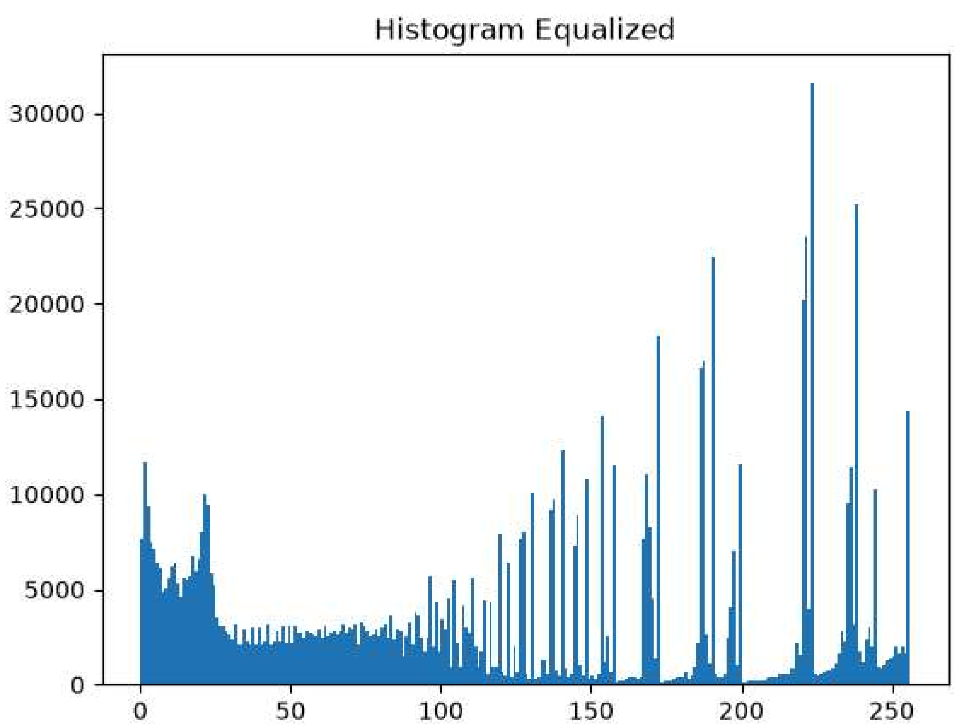
Output

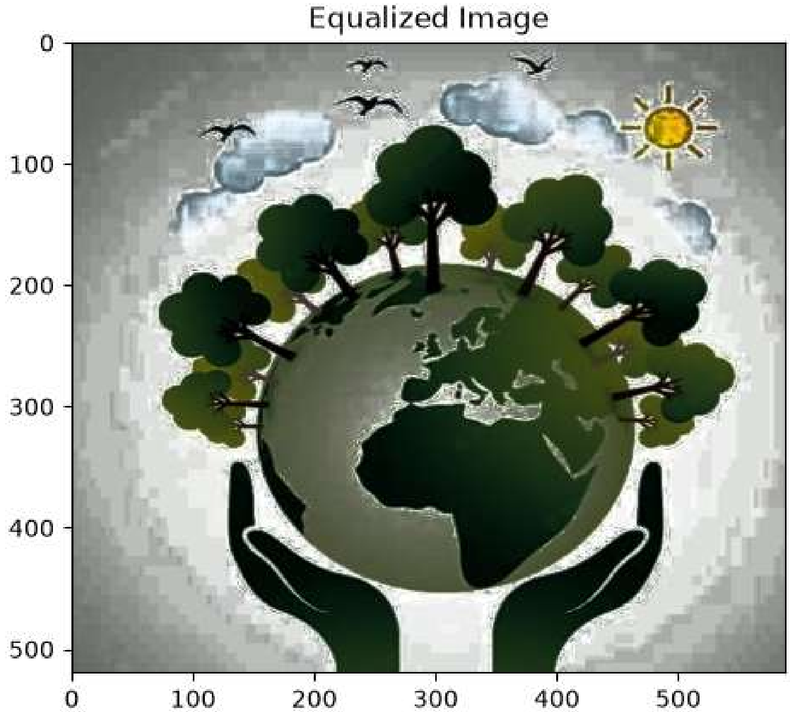


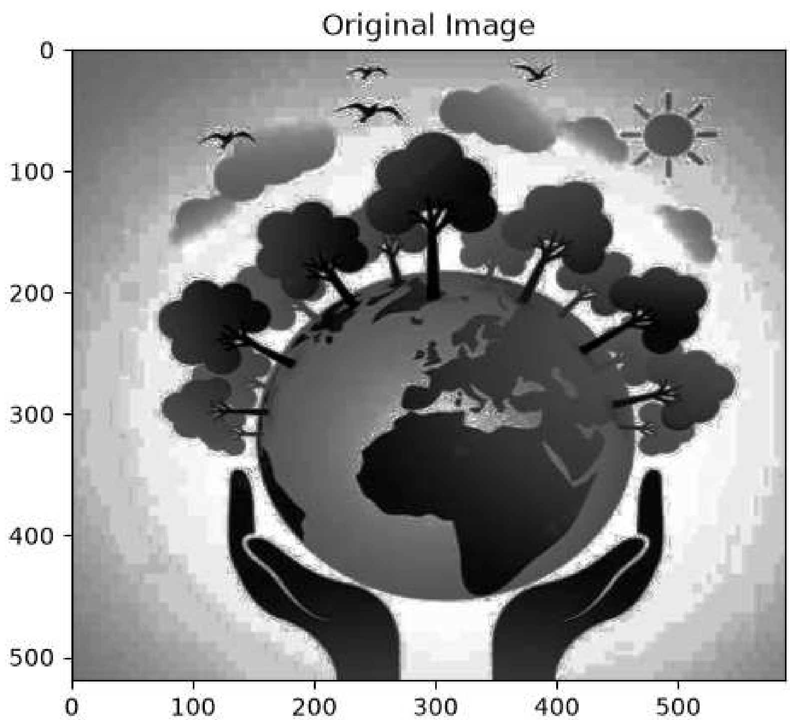


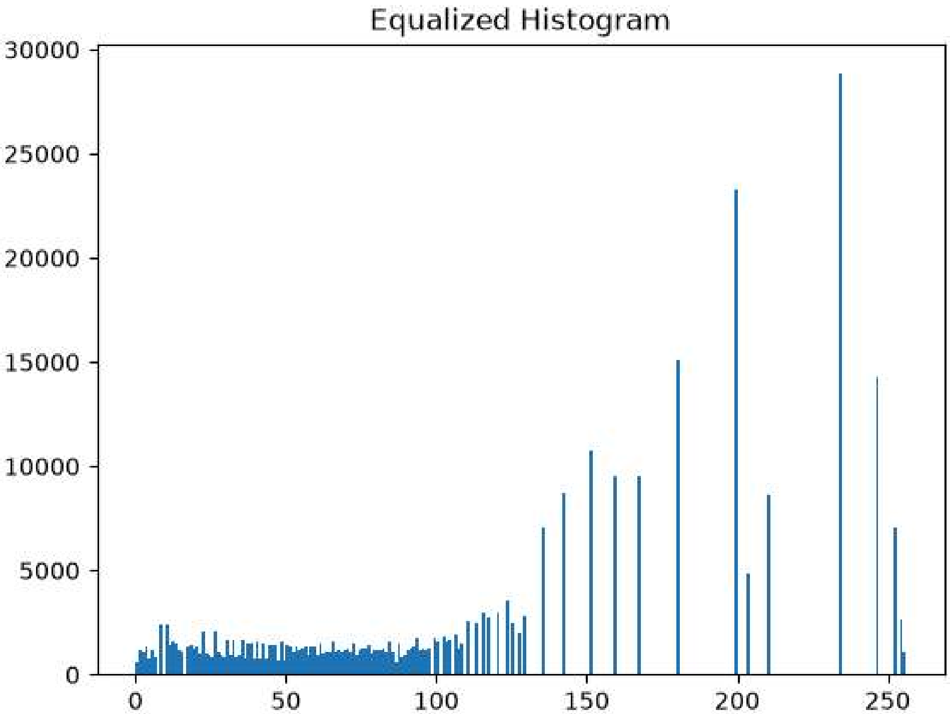


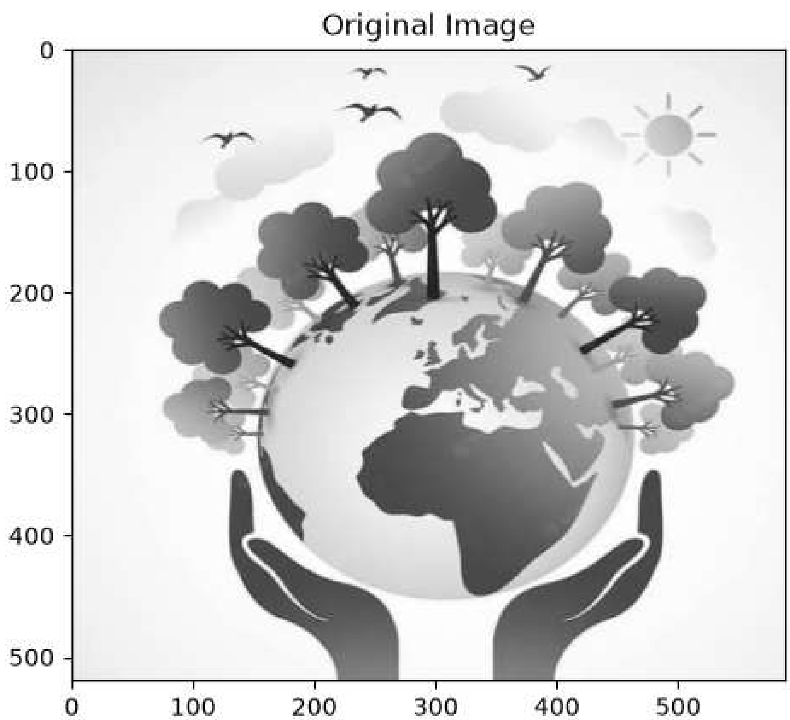


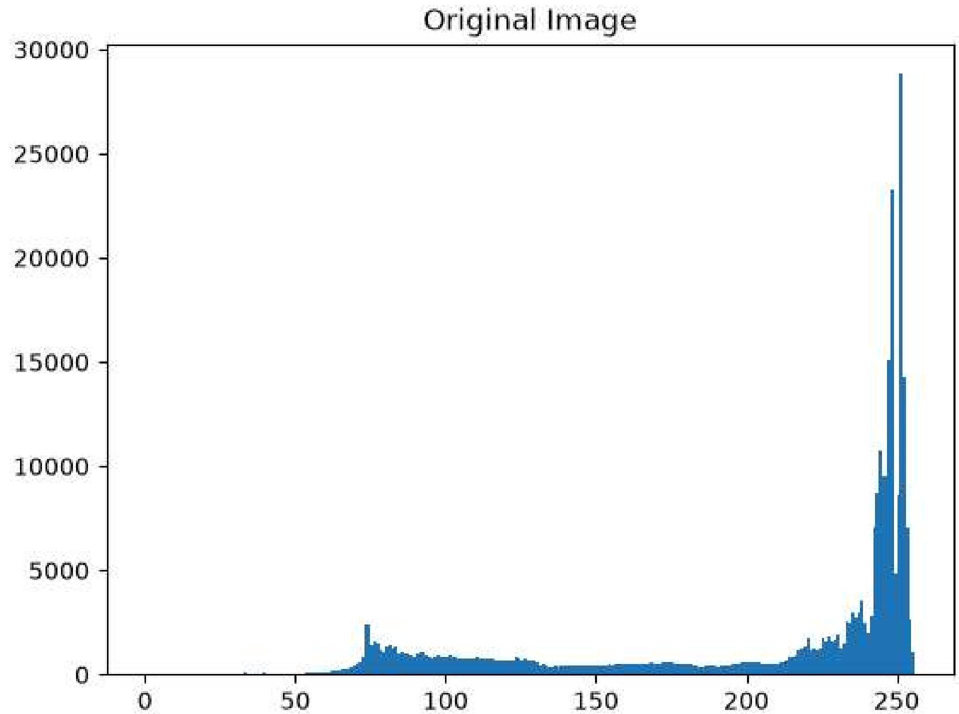



Equalized Image
0
100
200
300
400
500
0
100
200
300
400
500


Original Image
0
100
200
300
400
500
0
100
200
300
400
500


Equalized Histogram
30000
25000
20000
15000
10000
5000
0
0
50
100
150
200
250


Original Image
0
100
200
300
400
500
0
100
200
300
400
500



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Languages

Jupyter Notebook 100%